Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

Anonymous Authors

Abstract—**Artificial-intelligence (AI) data centers can provide spatial–temporal demand response (DR), yet a meter-only baseline cannot establish whether a claimed reduction is supported by committed work or displaced by rebound. This paper develops an auditable verifier that couples scheduler–telemetry provenance to workload-feasible counterfactuals, security-aware network value, and settlement. A sparse cumulative linear program (LP) enforces release, deadline, conservation, capacity, migration, and cross-midnight completion; an independent job-indexed LP checks all 71,128 positive-energy jobs. A validation-only risk envelope and a pre-event information gate separate contract formation from locked execution truth, while signed nodal value and preventive N–1 certificates bound payment over three telemetry-derived conversion scenarios. On a 121-day trace-driven inventory, 16 earlier days select parameters and 54 later days remain locked. The complete-ledger verifier achieves normalized root-mean-square error (nRMSE) 0.343 with 1.717 MWh/day of unsupported gross credit; at a one-hour gate, the committed-ledger replay achieves nRMSE 0.358, 3.442 MWh/day gross false credit, and 7.015 MWh/day meter-capped payable credit. The job-indexed witness spans 5,465,157 admissible job–slot variables, has maximum energy residual 2.17×10^{-19} MWh, and yields a 0.167-MWh net event reduction. A four-network alternating-current (AC) panel evaluates 9,648 native-connected outage cells with no voltage violation and maximum loading 0.999999996. Exact scale analysis reaches 3475.108× at the capacity-safe limit and 1.036× with the fixed 0.001-MW/GPU nameplate. The BurstGPT, MIT, and IEEE sources therefore form an auditable benchmark for security-aware DR verification under an observational, nonintervention setting.**

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. Introduction

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch graphics-processing-unit (GPU) work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value.

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice under published market guidance [1], despite known bias and manipulation incentives [2]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [2]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide the missing physical state. Temporal deferral, dynamic capacity, and geographical load balancing can reduce energy cost and peak demand [3]. Geographical assignment provides the complementary spatial control mechanism [4]. Recent work assesses data-driven flexibility from production traces and system-level security-constrained coordination, while balancing-market and clean-computing studies optimize spatial–temporal data-center dispatch [5], [6]. A recent power-system review also identifies quality-of-service (QoS), service-level-agreement (SLA), and market-participation constraints that are absent from meter-only accounting [7]. Hierarchical optimal-consumption models further connect facility efficiency with response quality. Virtual-link market clearing and space–time remuneration establish how geographical and temporal transfers can enter electricity markets [8], [9]. Receding-horizon operation and uncertainty-aware frequency-regulation coordination further establish the need to retain future arrivals and multi-site state [10]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains unresolved. This separation between computing feasibility and market verification motivates the auditable ledger used here. Recent whole-facility measurements likewise require a separate bottom-up conversion from GPU power to facility demand; the present study therefore keeps raw energy calibration, benchmark scaling, and network placement as separate evidence layers.

The second gap is electrical. Geographic workload control is commonly valued with energy prices or gross megawatts. Such accounting ignores the fact that nodal marginal value changes when congestion or a contingency constraint binds. Direct-current (DC) power flow and linear distribution factors provide transparent network sensitivities [11], [12]; N–1 dispatch can impose every credible post-contingency line limit through line-outage distribution factors (LODFs) [13]. No existing data-center DR verification framework jointly certifies workload feasibility, preserves signed spatial response, and settles the resulting security-aware network value. Alternating-current (AC) feasibility is a distinct nonlinear check, not a consequence of the DC certificate. Public workload and facility telemetry can support an observational replay of that ledger, but it does not constitute a causal utility event; a rigorous study must therefore separate measured-execution replay, counterfactual construction, and network-value verification to avoid conflating them with one field intervention.

This paper makes three power-system contributions. 1) We formulate a provenance-bound workload-state demand model in which release, deadline, conservation, cross-midnight completion, capacity, and migration constraints are enforced simultaneously. The counterfactual uses an exact aggregate energy-flow representation and a second globally solved job-indexed linear program (LP); the same immutable ledger is additionally replayed with measured contiguous job intervals, runtime, GPU count, and native power so that task-level feasibility is not inferred from a divisible-flow residual. 2) We construct a validation-only, continuous-arrival counterfactual verifier with a pointwise credit envelope. The output is an attainable spatial–temporal load trajectory with constraints absent from an unconstrained meter-only prediction, and its statistical performance is evaluated under an explicit false-credit, under-credit, and contract-overpayment audit. 3) We develop signed space–time nodal remuneration and a bilateral net-value contract under preventive N–1 dispatch, and map the feasible workload set into a payment uncertainty interval over held-out workload-to-power conversion. The central result is a coupling invariant that carries one committed workload state into physical verification, network valuation, and a set-valued settlement rule; the individual LP, dispatch, and allocation primitives are standard and are not claimed as new in isolation. Since public traces expose no utility geography, the spatial result is identified by a feature-stratified trace scenario and the complete permutation/penetration panel, not by an invented physical location. The certificate is conditional on trusted scheduler and telemetry provenance: a cryptographic digest verifies record integrity after commitment but does not authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. System Model and Problem Formulation

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The measured ledger supplies arrival energy a_{skt} for source s, class k, and release interval t. The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ. Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^{t} \sum_{d \in \mathcal{D}} x_{skd\tau}. \tag{1}$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^{t} a_{sk\tau}$ and deadline D_k, release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \tag{2}$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \tag{3}$$

The state recursion equals service, terminal service equals total arrivals, and $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \tag{4}$$

For rolling window end T^+ and final event-day release T^0, the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk,\max\{T^0, T^+ - D_k\}}, \tag{5}$$

which completes every job due inside the window and every event-day arrival, while retaining later real arrivals in the release constraints. The window starts $D_{\max} + 96$ intervals before the event day and ends $D_{\max}$ intervals after it; omitted earlier work must therefore expire before the evaluated day. Equations (2) and (3) prevent early execution and late completion, while (4) maps the feasible service state to the facility meter. The variable x is an aggregate, preemptive service envelope: it is valid for checkpointable batch work and does not by itself assert a nonpreemptive placement for every GPU job. For the measured MIT replay, each joined job is also retained with its contiguous execution interval, runtime, GPU count, and native average power; Experiment 14 checks that witness independently. This two-level construction keeps counterfactual optimization sparse while making the task-level scope explicit. For the counterfactual task-level check, let $j \in \mathcal{J}$ index a joined job, r_j denote its submitted release, and let d_j be the service-window endpoint formed from the submit-time scheduler "timelimit". The unlimited Slurm sentinel is mapped to the precommitted horizon $H_\infty = 128$ slots; finite declarations are retained exactly at their declared duration. The 0.001-MW-per-GPU nameplate is an audit-only feasibility precheck: if a declared window cannot serve E_j, the run fails closed and the window is not enlarged. Thus d_j is available at submission and never uses the observed completion time. Let E_j be measured energy, s_j its native region, and g_j its measured GPU count. The indexed service variable $u_{j\tau}$ is constrained by

$$\begin{aligned} & u_{j\tau} \geq 0, \qquad \sum_{\tau=r_j}^{d_j-1} u_{j\tau} = E_j, \\ & u_{j\tau} \leq g_j \bar{p}_{\text{GPU}} \Delta t, \\ & \sum_{j:s_j=d,\ r_j \leq \tau < d_j} u_{j\tau} \leq \bar{P}_d \Delta t, \\ & \min_u \sum_{j,\tau} \left[w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\} \right] u_{j\tau}. \end{aligned} \tag{6}$$

Here $\bar{p}_{\text{GPU}}$ is the precommitted 0.001 MW/GPU nameplate. Thus the counterfactual is a globally solved, job-indexed preemptive batch schedule with explicit release, submit-time service window, energy, site capacity, and GPU-count constraints. It is not obtained by replaying the measured profile; its event tariff is fixed before scoring. Experiment 19 solves (6) for all 71,128 positive-energy jobs. The observed $[t_j^{\text{start}}, t_j^{\text{end}}]$ interval is retained only to build the independent native replay used for scoring and never constrains $u_{j\tau}$. This model follows workload deferral and geographical assignment foundations [3], [4]; its rolling implementation follows the state-retention principle in [10]. Waiting and migration costs are $w_k(\tau - t) x_{skd\tau}$ and $m_{sd} x_{skd\tau}$, respectively. Because arrivals are fixed, replacing waiting duration by service time adds only a constant.

For provenance, each retained job is represented by the canonical tuple and its digest in

$$\begin{aligned} \ell_j &= (\text{id}_j, t_j^{\text{submit}}, t_j^{\text{start}}, t_j^{\text{end}}, E_j, n_j^{\text{tel}}, q_j, \kappa_j, \sigma_j), \\ \mathcal{L} &= \text{sort}\{\ell_j : j \in \mathcal{J}\}, \\ h_{\mathcal{L}} &= \text{SHA256}(\text{serialize}(\mathcal{L}))\,. \end{aligned} \tag{7}$$

where E_j is the positive aggregated Data Center GPU Manager (DCGM) energy and n_j^{tel} is its raw telemetry-record count. The digest uses the Secure Hash Standard [14]; the canonical row order, one-to-one join, release–execution order, and raw-to-join energy equality are checked independently of the scheduling linear program (LP). Consequently, the flow witness certifies the exact committed ledger and measured profile, while the digest does not make an unverified claim about jobs that might have been submitted before the commitment. Experiment 16 reports this provenance certificate and reconciles the measured regional capacity envelope with the 118-MW nameplate that was committed before the validation/test split; the envelope is not used to select that nameplate.

The trust boundary is explicit. The verifier assumes that the scheduler and DCGM services are the authoritative sources for the fields used in ℓ_j, and that the digest is committed before the event decision. Under this assumption, a changed record is detectable and a joined record cannot be silently replaced after commitment. The digest does not, however, establish that a semantically false deadline, region label, service class, or energy value was truthful when it was signed. Authenticity of those fields requires an upstream scheduler attestation, meter signature, or trusted execution interface; it is outside the optimization certificate proved here.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p^0_{dt} denotes the no-event counterfactual and $p^{1,\text{sim}}_{dt}$ denotes the event trajectory generated by the declared workload optimization. Experiment 13 and the trace-meter replay use p^{obs}_{dt} from the independently measured MIT/DCGM execution intervals. The public releases contain no utility event label, so p^{obs} is an observational alignment target and is never relabelled as a causal event outcome or supplied to an estimator before its decision. For mechanism-isolation rows, $r_{dt} = p^0_{dt} - p^{1,\text{sim}}_{dt}$ is signed: gross local response is $[r_{dt}]_+$, while net response retains negative rebound. For an estimate $\widehat{p}^0$, define the source-specific false credit against each trajectory:

$$F^{(z)} = \Delta t \sum_{d,t \in \mathcal{E}} [\widehat{p}^0_{dt} - p^{(z)}_{dt}]_+, \qquad z \in \{\text{obs}, \text{sim}\}. \tag{8}$$

The observed score uses the closed trace meter, whereas the mechanism-isolation score uses $p^{(\text{sim})} = p^{1,\text{sim}}$ from the response LP. Neither is a utility-event outcome. The offline fit uses the predeclared union ceiling $c^{\text{union}}_{dt} = \max\{p^{(\text{obs})}_{dt}, p^{(\text{sim})}_{dt}\}$ only as an eligibility screen; the audit recomputes (8) separately by day and source. The gross forecast credit is not itself a transfer. Before the event, the operator freezes a contract baseline p^{con}; after the event meter is closed, the payable response is the pointwise intersection

$$Q^{\text{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} \min\left\{[\widehat{p}^0_{dt} - p^{\text{obs}}_{dt}]_+, [p^{\text{con}}_{dt} - p^{\text{obs}}_{dt}]_+\right\}. \tag{9}$$

Equation (9) uses only the submitted baseline, the frozen contract baseline, and the closed event meter. The trace-anchored profile p^0 is retained exclusively as an offline diagnostic for oracle overpayment and underpayment; it never enters target fitting, gate decisions, workload optimization, or the contractual transfer. Thus the audit can expose a contract that overstates response without claiming that a post-event oracle is available to a deployed verifier. We report gross credit, contract-capped payable credit, oracle overpayment, oracle underpayment, normalized root-mean-square error (nRMSE), bias, precision, recall, and the F1 score (F_1).

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q, and DR price π^{DR}. If reference day j adds event-window service δR_j, the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^{H} \delta R_j - \sum_{j=1}^{H} \Delta C_j. \tag{10}$$

Let c'_j be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day j. Linear-program sensitivity [15] gives the following exact local condition:

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \tag{11}$$

At a degenerate optimum the dual at the current service level can be zero even when the right-direction marginal cost is positive. Experiment 1 therefore certifies c'_j with two independent right finite differences of 5×10^{-4} and 10^{-3} MWh and a 10^{-6} USD/MWh agreement tolerance. The historical-baseline premise is documented in [2]; (10)–(11) are this paper's workload-specific specialization.

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p^{\text{native}}_t + Mp^{\text{dc}}_t$ the nodal demand vector. For generator-to-bus map C_g, generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \ & \sum_i C_i(g_i) \\ \text{s.t. } & \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \le g \le \bar{g}, \\ & -\bar{f} \le H(C_g g - p) \le \bar{f}. \end{aligned} \tag{12}$$

The incidence mapping aggregates device demand at its connection bus, and (12) is the standard lossless DC representation [11], [12]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \tag{13}$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. Proposed Methodology

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations in Section II are the first stage of the methodology: they define the feasible workload state before any statistical target or payment is formed. Fig. 1 shows the complete chain. Meter history and the submitted-job ledger enter a statistical pre-estimator and the workload feasible set in parallel. The pre-estimator never receives execution energy used as scoring truth. For each event day, it uses only the information available under one of two declared protocols: lagged meter/calendar features for online baselines, or the complete submitted ledger for equally informed post-event comparisons. Ridge regression and gradient boosting are used as standard matched-information comparators [16], [17]; extremely randomized trees and synthetic control provide two additional nonlinear and donor-pool comparators [18], [19]. Two stronger comparators are also solved: median-loss gradient boosting with the complete submitted ledger and a nonnegative synthetic control whose weights globally minimize pre-event absolute error over 28 donor days. To separate the effect of projection from that of pre-estimation, the complete-ledger quantile output is also passed through the same exact workload-feasible projection, with its penalty chosen by contiguous validation folds. The same complete ledger is supplied to the post-event learner, the selected single projection, and the proposed convex verifier.

B. Sparse cumulative workload-state verifier

For statistical estimate $\widetilde{p}$ and each predeclared penalty ρ_ℓ, the first-stage program solves

$$\min_{x,y,z} \ \mathcal{C}_{\text{w}}(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \ge \pm(p_{dt}(x) - \widetilde{p}_{dt}), \tag{14}$$

subject to (1)–(4). Absolute-value epigraphs are standard convex constructions [15]; using them to project a meter baseline onto an auditable workload ledger is specific to this work. Here $\mathcal{C}_{\text{w}}$ denotes workload operating cost; C_i denotes generator-segment cost in (12), and $V(p)$ denotes the network dispatch value. Thus (14) changes the objective, not the physical feasible

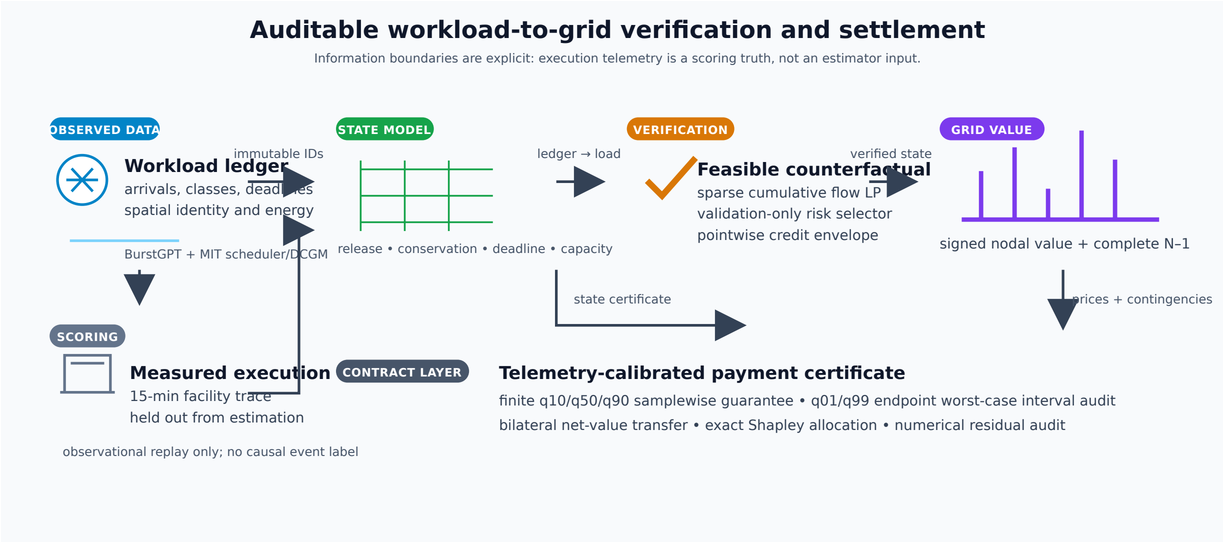

Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N–1 value and exact participant shares.

set. Six globally solved schedules $p^{(\ell)}$ share identical arrivals and constraints. Their validation-only convex combination $\bar{p} = \sum_\ell \alpha_\ell p^{(\ell)}$, $\alpha \geq 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The coefficients minimize validation squared error subject to both total and daily-tail false-credit budgets, where conditional value at risk (CVaR) measures the latter [20]. Let $\ell^\star$ be the single candidate with minimum worst contiguous-fold nRMSE and B_j^{cap} its validation false exposure on day j. For reserve $\beta \in (0, 1]$,

$$\sum_j F_j(\alpha) \leq \beta \sum_j B_j^{\mathrm{cap}}, \tag{15}$$

$$\mathrm{CVaR}_{0.75}(F_j(\alpha)) \leq \beta \mathrm{CVaR}_{0.75}(B_j^{\mathrm{cap}}). \tag{16}$$

For implementation, the positive part is not differentiated through a sort. For each event sample n and validation day j, the program introduces $f_n \geq 0$, a CVaR threshold $\nu \geq 0$, and tail slacks $u_j \geq 0$:

$$f_n \geq p_n^\alpha - c_n^{\mathrm{union}}, \tag{17}$$

$$F_j = \sum_{n \in j} f_n, \tag{18}$$

$$F_j - \nu - u_j \leq 0, \tag{19}$$

$$\nu + \frac{1}{0.25J} \sum_j u_j \leq \beta\, \mathrm{CVaR}_{0.75}(B_j^{\mathrm{cap}}). \tag{20}$$

Here c^{union} is the explicitly labelled offline risk-selection ceiling formed from the two separate trajectories. It is never substituted for the frozen contract profile p^{con} in (9). The explicit epigraph in (20) is the implementation used in all reported fits. Together with the simplex equality, box bounds on α, and the total budget in (15), these are linear constraints and the squared error objective is a convex quadratic program. HiGHS first supplies a feasible epigraph point; the full epigraph is then solved with a sparse primal–dual trust-region method. The returned stationarity residual, every linear epigraph residual, the simplex equality, and all bound residuals are recomputed from the final primal point. The optimizer status and these residuals are stored for every fit. Four contiguous folds select β using validation nRMSE, then risk ratio; held-out noninferiority is reported as an outcome and is not used as a hidden constraint.

A second exact projection targets $\bar{p}$ while imposing, for every event sample,

$$p_{dt}^{\mathrm{safe}} \leq p_{dt}^{(\ell^\star)}. \tag{21}$$

The pointwise cap in (21) is the upper side of the contractual credit band; it is evaluated on every event slot before settlement.

a) Two-sided credit band.: The upper envelope alone would admit an arbitrarily conservative zero-credit profile. We therefore impose a lower band tied to the independently fitted convex target. Define the componentwise risk floor $p_{dt}^{\mathrm{floor}} = \min\{p_{dt}^{(\ell^\star)}, \bar{p}_{dt}\}$, where $\bar{p}$ is the validation-fitted convex target before the final workload projection. The event-slot constraints are

$$p_{dt}^{\mathrm{floor}} - \varepsilon \leq p_{dt}^{\mathrm{safe}} \leq p_{dt}^{(\ell^\star)}, \qquad (d, t) \in \mathcal{D} \times \mathcal{E}, \tag{22}$$

where $\varepsilon = 1$ MW is fixed before the locked test split. The lower bound is implemented together with the workload constraints, not as a post-processing clip. Thus the output remains an attainable schedule and can move below the single reference when the independent risk fit supports a safer trajectory. The total constraint (15) controls aggregate exposure and (16) controls the worst daily tail. The upper side is the deterministic payment cap; the lower side is a declared under-credit regularizer relative to p^{floor}, not a distribution-free guarantee against an arbitrary unseen meter trace. Overpayment against the realized event is handled by the contract-capped settlement rule (9). The selected single feasible projection proves nonemptiness, and the locked results report both margins and the post-event payable amount.

b) Decision-time information boundary.: The ex-post selector can use the complete submitted ledger because it is an offline settlement protocol. For an event-gate deployment, all arrivals after the gate are removed before both the gate-consistent target construction and optimization. The target is refit with the same metadata gradient-boosting specification used by the matched-information baseline, using historical days unchanged and the current day's masked ledger for every feature row. A first exact LP freezes the gate-consistent no-event profile as the contract baseline. A second LP uses the same masked ledger and the declared DR price on the participating event slots to shift eligible service away from the event window, while closing the rolling checkpoint at the end of the event day; unexpired batch state is carried to the next checkpoint.

It never credits a reserved future job. After the event, its gross forecast credit is converted to the payable quantity in (9); the resulting payable response is therefore generated by an explicit workload-feasible response objective without suppressing the service trajectory or using future arrivals. The rolling-service response is reported separately from the frozen contract profile, information-boundary diagnostic, and complete-ledger benchmark.

To avoid conflating capacity planning with payment eligibility, the gate also publishes a gate-consistent reserve envelope. For a candidate quantile η and event day d, the reserve for a post-gate release cell is defined by (23):

$$\widehat{a}^{\mathrm{res}}_{sk\tau}(d;\eta) = Q_\eta(\{a_{sk\tau}(h) : h < d\}), \qquad \tau \geq t_{\mathrm{gate}}, \quad (23)$$

where the quantile is computed from temporally earlier days only. The finite candidate set is selected on validation days under a declared false-credit budget, and the resulting reserve is used for capacity planning only. The payable event profile remains the committed-ledger LP above; no reserved job can enter the payment numerator before its later ledger commitment. This separation supplies an information-boundary forecast without silently reintroducing future arrivals into the certificate.

The cumulative state reduces the constraint-matrix nonzeros from $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|^2)$ for explicit prefixes to $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$. A 512-slot deadline is handled by (5) with every observed future arrival retained; no zero-arrival completion buffer is used in the final continuous-horizon certificate. The day trajectory is embedded by a lexicographic pair of linear programs: minimum L1 distance first, followed by minimum operating cost on that optimal-distance face. Hence no penalty weight trades physical fit against cost. All linear programs are solved to global optimality with HiGHS [21]. The resulting state-to-credit sequence is summarized in Fig. 2; the execution replay and interval audit are reported separately in Experiments 13–15.

C. Signed nodal and N–1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top(p^0 - p^1)$, while (13) is the realized system-value benchmark and the payoff of an explicit bilateral avoided-cost contract; it is not relabeled as an independent system operator (ISO) market payment. The convex subgradient inequality [15] yields the global certificate

$$0 \leq (\lambda^0)^\top(p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top(p^1 - p^0). \quad (24)$$

Thus the linear rule globally upper-bounds exact value in the implemented polyhedral market model. Equality holds only when both endpoints share a supporting dual subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from the earliest deadline-feasible anticipatory shift through the latest recovery interval. The space–time rule is

$$\Pi^{\mathrm{ST}} = \Delta t \sum_{t\in\mathcal{W}} \sum_{d} \lambda^0_{dt}(p^0_{dt} - p^1_{dt}). \quad (25)$$

This is the nodal virtual-link remuneration structure of [8], [9], applied here to a verified counterfactual. Summing site payments reproduces (25) exactly.

Space–time energy remuneration is only one component of the contracted DR product. Let $\mathcal{D}^{\mathrm{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\mathrm{cap}} = \Delta t \sum_{d\in\mathcal{D}^{\mathrm{part}}} \sum_{t\in\mathcal{E}} (p^{\emptyset}_{dt} - p^1_{dt}), \quad (26)$$

$$G = r^{\mathrm{DR}} Q^{\mathrm{cap}} + \Pi^{\mathrm{ST}}, \quad (27)$$

where $p^{\emptyset}$ is the minimum-cost no-event schedule under the identical continuous arrival window. Capacity credits and ex-post participation constraints follow established DR mechanism design [22], [23]; the signed Π^{ST} term separately debits spatial transfer and recovery. Equations (26)–(27) jointly define the signed service and operator value.

Let x^1 and $x^{\emptyset}$ denote the workload service schedules inducing p^1 and $p^{\emptyset}$, respectively. Define $C_{\mathrm{opp}} = \mathcal{C}_{\mathrm{w}}(x^1) - \mathcal{C}_{\mathrm{w}}(x^{\emptyset})$ as the participant opportunity cost and $S = G - C_{\mathrm{opp}}$ as the transferable transaction surplus. With disagreement utilities equal to zero, the symmetric Nash solution [24] activates the contract iff $S \geq 0$ and sets

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\mathrm{B}} = z\left(C_{\mathrm{opp}} + \frac{S}{2}\right), \quad u_{\mathrm{dc}} = u_{\mathrm{op}} = z\frac{S}{2}. \quad (28)$$

Thus (28) makes voluntary nonactivation the explicit outside option. For every activated contract, both utilities are nonnegative and the single transfer P^{B} is exactly budget balanced. The bilateral transfer is reported separately from both ISO nodal remuneration and the exact dispatch value benchmark. Unlike event-only accounting, $\mathcal{W}$ debits both pre-event load increases and delayed rebound; workload conservation gives $\sum_{d,t\in\mathcal{W}}(p^0_{dt} - p^1_{dt})\Delta t = 0$ up to solver tolerance.

Security-aware settlement augments (12) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ, then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k} H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \quad (29)$$

Equation (29) is the standard preventive N–1 LODF formulation [13]; the experiment imposes all credible rows simultaneously, with no contingency screened out. Native public ratings are retained. Islanding outages are reported and excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the scope of DC security models and the N–1 planning criterion [25].

D. Scenario-robust N–1 payment certificate

The pointwise envelope controls credited energy but does not by itself certify network payment because spatially different demand vectors can activate different security constraints. Let $\mathcal{P} = \mathrm{conv}\{p^{(1)}, \dots, p^{(L)}\}$ be the convex hull of complete workload-feasible schedules and let $p^{\mathrm{cap}} \in \mathcal{P}$ be the validation-selected single feasible projection. Its penalty $\rho_{\ell^\star}$ is selected by contiguous validation and is reported separately from the payment-target penalty. The feasible-quantile projection is retained as an external matched transfer comparator in Section V. Let Ξ_3 contain the 10th, 50th, and 90th percentiles of held-out per-job measured-to-predicted GPU energy. These predeclared empirical scenarios define the finite power-conversion contract supported by independent telemetry [26]. For each settlement day, the payment-certified verifier defines

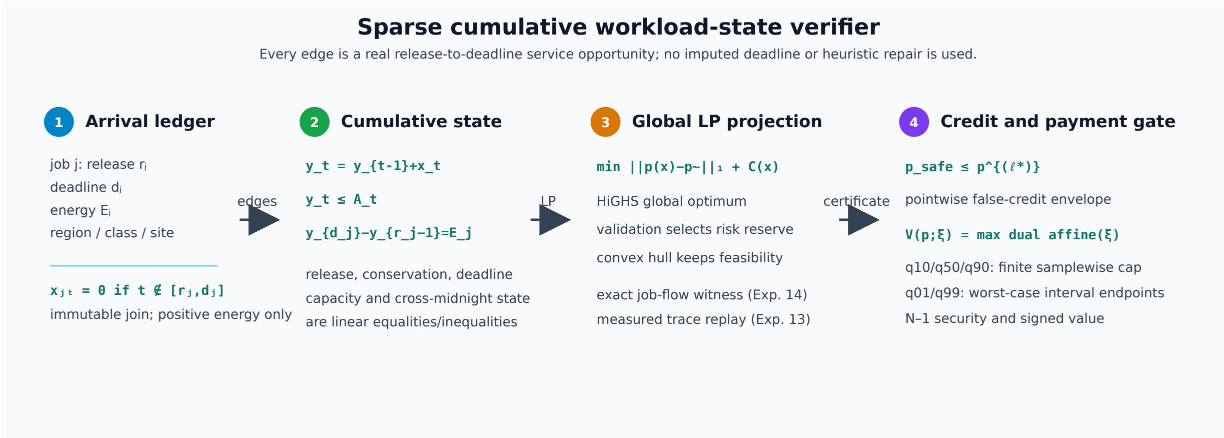

Fig. 2. Sparse cumulative workload-state verifier. Real release-to-deadline edges feed the global feasibility LP; the pointwise credit envelope and the finite/interval payment gates are applied only after the workload state is certified.

$B_\xi^{\mathrm{cap}} = \Delta t \sum_{t\in\mathcal{E}} V_{N-1}(p_t^{\mathrm{native}} + M\xi p_t^{\mathrm{cap}})$ and solves

$$\begin{aligned}
&\operatorname{lexmin}_{\alpha,e,\eta,\{g_t^\xi\}} \quad \left(\sum_{d,t\in\mathcal{E}} e_{dt}, -\eta\right) \qquad (30)\\
&\text{s.t.} \quad p^\alpha = \sum_{\ell=1}^{L} \alpha_\ell p^{(\ell)},\\
&\qquad \alpha \ge 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \le \eta \le 1,\\
&\qquad e_{dt} \ge \pm(p_{dt}^\alpha - p_{dt}^{\mathrm{safe}}),\\
&\qquad g_t^\xi \text{ satisfies (12) and (29)},\\
&\qquad\quad \text{with } p_t = p_t^{\mathrm{native}} + M\xi p_t^\alpha,\\
&\qquad\qquad t\in\mathcal{E}, \quad \xi \in \Xi_3,\\
&\qquad \Delta t \sum_{t\in\mathcal{E}} \sum_i C_i(g_{it}^\xi)\\
&\qquad\quad \le (1-\eta) B_\xi^{\mathrm{cap}}, \quad \xi\in\Xi_3.
\end{aligned}$$

Generator costs use the same declared piecewise-linear epigraph as market settlement. Two global linear programs implement the lexicographic objective: the first obtains the minimum trajectory error, and the second preserves that optimum within solver tolerance while maximizing the common fractional margin η. Existence of the primal dispatches in (30) is equivalent to the corresponding upper bound on the optimal N–1 value; the unit weight on p^{cap} proves nonemptiness in every scenario. Consequently, for any $\xi \in \Xi_3$ and realized event load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned}
&\sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p_t^{\mathrm{native}} + M\xi p_t^\alpha) - V_{N-1}(p_t^{1,\xi})]\\
&\le \sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p_t^{\mathrm{native}} + M\xi p_t^{\mathrm{cap}}) - V_{N-1}(p_t^{1,\xi})]. \qquad (31)
\end{aligned}$$

The common realized-cost term cancels, so the certificate requires no execution truth and holds samplewise. Both programs contain every finite non-islanding contingency. The formulation follows the standard SCED primal and linear-program epigraph foundations [12], [15]. Here $\mathcal{P}$ contains the six independently solved first-stage projections from (14) and the matched feasible-quantile projection. The risk-constrained verifier is an external target in the absolute-deviation objective and is not a selectable candidate; the quantile schedule is used only as an independent transfer comparator, not as the contractual payment cap.

The finite cap is complemented by a set-valued payment certificate. The contractual cap profile is denoted $p^{\mathrm{cap}} = p^{(\ell^\star)}$, where $\ell^\star$ is selected by contiguous validation nRMSE; the independent payment-target profile p^{pay} is selected separately by validation payment error and is used only as the target of the certificate. For a realized counterfactual value, define $\Delta V_\xi(p) = V_{N-1}(p,\xi) - V_{N-1}(p^{1,\xi})$ and the continuous segment $p_\theta = (1-\theta)p^{\mathrm{cap}} + \theta p^{\mathrm{cert}}$, $0 \le \theta \le 1$. The induced interval is

$$\mathcal{I}_\xi = \left[\min_{0\le\theta\le 1} \Delta V_\xi(p_\theta),\ \max\{\Delta V_\xi(p^{\mathrm{cap}}), \Delta V_\xi(p^{\mathrm{cert}})\}\right]. \qquad (32)$$

The lower endpoint is obtained by one exact joint N–1 SCED LP with a shared segment parameter; the upper endpoint follows from convexity of the optimal dispatch value. The selected cap payment is an element of $\mathcal{I}_\xi$. The endpoints use only validation-frozen feasible profiles and the realized counterfactual needed for settlement; no oracle baseline selects or certifies the interval. Interval width is reported as uncertainty cost, while the seven-profile hull remains the finite-scenario payment certificate in (30).

E. Participant allocation

For participants N, define $v(S)$ as signed dispatch value when coalition S moves from baseline to actual demand and nonmembers remain at baseline. The exact allocation is

$$\phi_i(v) = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!}[v(S\cup\{i\}) - v(S)]. \qquad (33)$$

This is the Shapley value [27]; the characteristic function based on signed grid value is the proposed specialization. Grouping (33) by coalition size cancels every intermediate value and gives $\sum_i \phi_i = v(N)$ when $v(\varnothing) = 0$. We enumerate all 16 coalitions for four sites and all 256 coalitions for eight contractual portfolios; no permutation estimator is used. For larger auditable contracts, exchangeable slices at the same electrical site are grouped by their member count. The exact sum retains the binomial multiplicity of every labeled coalition and evaluates only $\prod_g (m_g + 1)$ distinct count states. This symmetry reduction remains the Shapley value and is evaluated through 20 participants.

F. Analytical guarantees

Proposition 1 (manipulation boundary): Under a ten-day arithmetic baseline, differentiable one-sided optimal costs, and an interior feasible increase in at least one reference-day event service, a strictly profitable local manipulation exists if and only if (11) holds. To prove the statement, differentiate (10) along any feasible reference-day direction. The marginal payment is the same $q\pi^{\mathrm{DR}}/H$ for every day, whereas the least physical cost is $\min_j c'_j$. A strict excess selects a positive direction on an argmin day; if no strict excess exists, every nonnegative local direction has nonpositive first-order profit. Equality admits indifference but does not imply strict profit. Experiment 1 computes c'_j from the honest program's right-hand-side dual and solves the strategic program independently, preventing the behavioral result from being reused as its own certificate.

Proposition 2 (workload and two-sided credit feasibility): For fixed arrivals, every convex combination of solutions to (1)–(4) is feasible, and the envelope output satisfies the deterministic two-sided band (22) on every event cell. Consequently, for every possible unseen execution c_{dt}, its false-credit energy is no larger than that of the selected cap $p^{(\ell^\star)}$, while its additional under-credit is bounded by $\varepsilon|\mathcal{D}||\mathcal{E}|\Delta t$ relative to the independent floor p^{floor}. Affine equalities are preserved by the simplex coefficients; each linear inequality is preserved by convexity. The upper inequality in (22) and monotonicity of $[z-c]_+$ give

$$[p_{dt}^{\mathrm{safe}} - c_{dt}]_+ \leq [p_{dt}^{(\ell^\star)} - c_{dt}]_+.$$

The lower inequality gives $[c_{dt} - p_{dt}^{\mathrm{safe}}]_+ \leq [c_{dt} - p_{dt}^{\mathrm{floor}}]_+ + \varepsilon$. Summation over sites and event intervals proves both bounds. The result is a feasibility-and-band certificate determined by the declared constraints and ε, not by the distribution of the locked test set.

Proposition 3 (security-aware value consistency): For the piecewise-linear base-case or N–1 dispatch, exact signed settlement equals the corresponding optimal-value difference and the nodal linear rule upper-bounds it by (24). The feasible-set right-hand side is affine in nodal demand, so the optimal value is convex and piecewise affine. Applying the subgradient inequality at p^0 gives (24); applying it again at p^1 yields

$$\begin{aligned} 0 \leq \Delta V_{\mathrm{lin}} - \Delta V &\leq (\lambda^1 - \lambda^0)^\top (p^1 - p^0) \\ &\leq \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|. \end{aligned}$$

Here $\|\cdot\|$ and $\|\cdot\|_*$ denote a fixed dual pair of norms on the nodal-demand and price spaces. The same proof applies after stacking all rows in (29), because they remain linear. Gross positive-only settlement has no analogous identity: truncation changes the response vector before it enters the value function.

Proposition 4 (robust daily payment noninferiority): Every solution of (30) is workload feasible and its exact N–1 payment is no larger than the contractual cap payment for every realized event trajectory and every declared $\xi \in \Xi_3$. Workload feasibility follows from Proposition 2. For each ξ, the embedded dispatch is a feasible witness whose cost satisfies the corresponding last constraint in (30); optimal N–1 dispatch can only have lower cost. Subtracting the same scenario-specific realized cost gives (31). Conditional on the telemetry-derived finite set Ξ_3, the proof is deterministic and contains no test-outcome assumption.

Proposition 5 (continuous worst-case interval certificate): For a fixed workload profile p, the optimal piecewise-linear N–1 dispatch value $V_{N-1}(p;\xi)$ is convex in a scalar conversion factor $\xi \in [\xi_{\mathrm{L}}, \xi_{\mathrm{U}}]$. Therefore

$$\max_{\xi \in [\xi_{\mathrm{L}}, \xi_{\mathrm{U}}]} V_{N-1}(p;\xi) = \max\{V_{N-1}(p;\xi_{\mathrm{L}}), V_{N-1}(p;\xi_{\mathrm{U}})\}.$$

The value function is the supremum of affine dual objectives in ξ, hence convex; a convex function on a compact interval attains its maximum at an endpoint. If the two endpoint inequalities $V_{N-1}(p^\alpha;\xi_e) \leq V_{N-1}(p^{\mathrm{cap}};\xi_e)$ hold for $e \in \{\mathrm{L},\mathrm{U}\}$, then the certified candidate also has no larger worst-case interval cost than the contractual cap. This interval result is a worst-case bound; pointwise dominance at every interior ξ is asserted only for the finite set Ξ_3 in Proposition 4. Experiment 15 evaluates the capacity-safe endpoints with the exact four-segment contractual N–1 model, while Experiment 9 independently replays Ξ_3 at 40 segments; the two resolutions are not conflated.

Proposition 6 (workload-segment payment interval): For a fixed conversion factor and realized counterfactual, the lower endpoint in (32) is the exact minimum payment over the declared continuous segment because the shared segment parameter and all interval-wise preventive N–1 dispatch variables are solved in one joint LP. The upper endpoint is the exact maximum because the optimal dispatch value is convex in the affine profile and a convex function attains its maximum on a compact segment at a vertex. Thus (32) is exact for the contractual segment, not a grid approximation or a claim that the segment contains every physically possible schedule. Experiment 15 reports the joint-LP minimum, endpoint maximum, and resulting width at both held-out conversion endpoints; no oracle baseline selects the segment.

Proposition 7 (provenance-bound workload witness): Suppose the canonical tuple in (7) is computed from the retained scheduler–DCGM join after the declared duplicate and positive-energy rules. If the certificate verifies one-to-one joining, release–execution ordering, raw-to-join energy conservation, and zero LP residuals, then the job-level flow witness is a feasible representation of that committed ledger. Under the collision-resistance assumption of SHA-256, any post-commit modification of a tuple or its canonical ordering changes the digest except with negligible probability. The statement binds the computational witness to the committed records; it does not certify that an uncommitted pre-event submission was truthful. Experiment 16 checks each premise independently and records the raw source hashes, so the provenance claim is separate from the scheduling solver's own feasibility status.

Proposition 8 (indexed counterfactual feasibility): If the job-indexed program (6) returns an optimal solution, every retained job receives exactly its committed energy within its submitted release–deadline window, every site-slot total respects the declared capacity, and each service interval respects the measured GPU-count power bound. These statements follow directly from the equality, bound, and capacity rows of (6); they do not rely on aggregate-flow rounding. The guarantee is for preemptive checkpointable service. Nonpreemptive contiguous execution is asserted only for the independently observed intervals audited in Experiment 14.

Proposition 9 (ledger-to-settlement coupling invariant): For an aggregate schedule x and a job-indexed witness u over the same committed ledger, let $X_{dt} = \Delta t^{-1} \sum_{s,k} x_{skdt}$ and use the identical conversion in $p = P^{\mathrm{fix}} + MX$. Both representations therefore induce the same nodal demand; signed value, payment caps, and Shapley allocations act on this one p, with $\sum_i \phi_i = v(N)$. The invariant requires the ledger, conversion, and baseline to remain committed between verification and settlement.

IV. Experimental Design

A. Data, preprocessing, and information boundary

Table I reports the complete data inventory. BurstGPT contributes 5,188,507 successful requests and observed token service [28]. MIT Supercloud contributes scheduler submissions and independent Data Center GPU Manager energy. The complete positive-energy scheduler–DCGM join contains 71,128 jobs, of which 68,664 start within the common 121-day tensor window used by Experiments 1–13 [26]. Submission time creates queue arrivals; measured execution intervals create scoring truth. A held-out GPU-power regression uses 28,413 observations and obtains $R^2 = 0.899$, mean absolute error

TABLE I
COMPLETE DATA AND EVALUATION PROTOCOL.

Item	Declared use and count
BurstGPT	5,188,507 requests; 2.176 billion tokens; inference arrivals and observed service
MIT scheduler	287,173 raw rows; 104 duplicate rows removed, leaving 287,069 canonical rows; submission time, declared timelimit, GPU count, and queue state
MIT GPU telemetry	96,893 rows; 68,664 tensor-window jobs and 71,128 full-horizon positive-energy jobs; measured execution energy
Time resolution	15 min; 121 calendar days; 121 valid complete days
Evaluation	16 validation days followed by 54 untouched test days
Networks	IEEE Reliability Test System 24-bus (RTS-24), IEEE-30, IEEE-39, and IEEE-118 for AC/N–1 panels; IEEE-300 is added in the resolution-convergence panel
Provenance	Immutable scheduler/DCGM join digest, raw-to-join energy conservation, and measured regional capacity envelope

(MAE) 11.22 W, and root-mean-square error (RMSE) 15.82 W. All 99th-percentile workload scaling factors are fitted on the first 40 complete days of the training portion; validation and locked-test days are excluded from that fit. Per-job measured-to-predicted energy ratios define Ξ_3 in (30); no synthetic row replaces an invalid record. The 68,664 count is the 121-day workload tensor used by Experiments 1–13. Experiment 14 does not truncate at that tensor boundary: it extends the observed replay horizon to the last scheduler completion and retains all 71,128 positive-energy joined jobs for its independent job-level flow witness. Experiment 19 uses the same join but replaces the observed completion interval with the submit-time timelimit window and a precommitted unlimited-window rule.

The public traces do not contain utility bus locations, so the four-region coupling is evaluated under a trace-driven benchmark setting. Regions are assigned by a predeclared feature-stratified round robin over class, GPU count, runtime, submission time, and energy. Experiment 11 exhausts all $4! = 24$ permutations, three penetrations, and two concentration controls. Experiment 18 uses fixed generator-bus electrical-role strata in each public case and the same 0.90 native-load multiplier as the primary preventive DC stress point; its cross-network result is therefore conditional on the declared trace-to-bus scenario, not evidence of utility co-location. Held-out energy-ratio factors span 0.5845–2.6931; the contract uses the capacity-safe upper factor 1.1499 because the raw upper endpoint exceeds the committed 118-MW nameplate. Public Power Grid Library (PGLib) and MATPOWER/PYPOWER cases supply topology, ratings, and costs [12], [29]; RTS-24 follows the published data [30].

B. Baselines, parameters, and inference

Experiment 2 compares eight meter/statistical baselines with the complete-ledger quantile, its exact feasible projection, pre-event synthetic control, tail-risk counterfactual, the selected single projection, and risk-constrained verifier. The payment-certified verifieris evaluated separately in Experiments 9 and 15 because its candidate hull and N–1 objective are distinct. An information-set audit verifies feature equality and excludes execution truth. Real-time, elastic, and batch deferral windows are 0, 4, and 512 slots; flexible capacity is 118 MW plus 6 MW fixed demand. Six projection penalties and four risk reserves are validation-only; nested contiguous-fold ordering selects the reserve before locked days are opened, and the four committed-response projection weights are calibrated separately on earlier validation days. Payments are scored by an independent 40-segment evaluator. Moving three-day bootstrap replicates and all 2^{18} block-sign assignments use Holm correction [31], [32]. The sparse solver grows from 5,760 variables/13,424 nonzeros at 96 slots to 36,480/85,488 at 608 slots, with 0.039–0.212-s recorded solve times.

The model-out check uses polar alternating-current optimal power flow (AC OPF) with active/reactive balance, voltage and reactive-generation bounds, and apparent-power ratings [12]. After the intact solve, every credible outage imposes, for each non-reference generator i,

$$P_{Gi}^{(k)} = P_{Gi}^{(0)}, \qquad i \notin \mathcal{G}_{\mathrm{ref}},\ k \in \mathcal{C}. \tag{34}$$

The reference generator balances outage-dependent active losses, while $Q_G^{(k)}$ and voltages remain recourse. Apparent-power and voltage limits are checked after each fixed-plan solve, so (34) tests one shared active plan; post- contingency active schedules are not reoptimized. The implementation sets the non-reference bounds to the intact output plus/minus 10^{-8} MW and accepts only a further 10^{-6}-MW solver-residual margin; the largest recorded deviation is 2.52×10^{-8} MW.

Experiments 1–2 test incentives and workload verification; 3–9 test signed settlement, network transfer, risk certification, and N–1 value; 10–12 test AC feasibility, spatial/scale transfer, and continuous-cycle remuneration. Experiments 13–16 provide measured replay, a job-level witness, conversion intervals, and immutable ledger reconciliation. Experiment 17 freezes the event-gate information set, Experiment 18 evaluates a fixed-active-plan AC certificate across four public networks, Experiment 19 solves the indexed job-level counterfactual, and Experiment 21 audits the exact homogeneous capacity-safe scaling of that primal. Confirmatory comparisons use the same 54 locked days unless a deterministic network or ledger rule is stated.

V. Results

A. Incentive boundary and independent verification

Experiment 1 solves the entire 8×7 DR-price/call-probability grid. The dual threshold (11) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 8.223 MWh. Of 55.452 MWh credited, 44.458 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.802. The profitable region follows the derived marginal boundary and does not use a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates all twelve methods on 54 later days in the mechanism-isolation response panel. Table II separates the validation-fitted tail-risk counterfactual ensemble from the final pointwise-safe output. Against the selected single projection, tail-risk counterfactualimproves F_1 from 0.442 to 0.484 and nRMSE from 0.345 to 0.331, but increases unsupported credit from 1.849 to 2.524 MWh/day. The final risk-constrained output has nRMSE 0.343, false credit 1.717 MWh/day, and $F_1 = 0.442$; its pointwise upper cap is therefore nondegenerate and its false-credit reduction relative to the single projection is 7.15%. Against the equally informed feasible-quantile projection, false credit decreases from 2.917 to 1.717 MWh/day, while nRMSE increases from 0.321 to 0.343. These results locate the verifier on an explicit risk–accuracy frontier. Source-separated locked-test false credit is 7.467 versus 7.617 MWh/day (final/single) for the observed meter and 10.534 versus 10.706 for the mechanism-isolation trajectory; both diagnostics remain observational.

The independent convex fit is not identified with the single projection. Relative to that reference, total-plus-CVaR, CVaR-only, and the final pointwise envelope change (nRMSE, false credit) by $(-0.014, +0.674)$, $(-0.014, +1.949)$, and $(-0.002, -0.132)$, respectively. Every locked profile passes

TABLE II
LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS).

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	4.216	172.826	6.423	0.143
Ridge	2.773	110.629	5.083	0.235
Gradient boosting	2.354	97.178	4.944	0.245
Extra Trees	3.338	133.129	6.610	0.175
Online metadata	2.136	90.619	4.171	0.265
Post-event metadata	2.127	91.641	4.091	0.268
Post-event quantile	0.946	24.156	6.087	0.537
Synthetic control	2.553	101.898	7.912	0.206
Feasible quantile	**0.321**	2.917	11.493	**0.508**
Single feasible	0.345	1.849	13.431	0.442
Tail-risk feasible	**0.331**	2.524	11.981	0.484
Risk-constrained safe	0.343	**1.717**	13.471	0.442

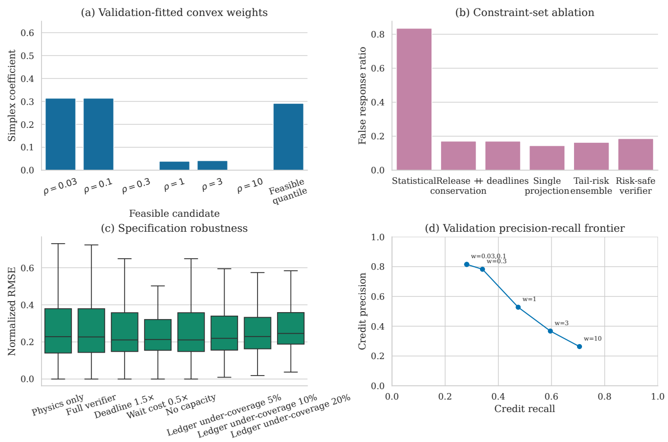

Fig. 3. Validation-fitted projection weights, constraint-set ablation, specification robustness, and the precision–recall frontier. Every point is produced by a globally solved workload program.

the simplex and two-sided residual checks. Removing both risk constraints gives 4.680 MWh/day false credit; total-only, CVaR-only, and combined constraints give 2.524, 3.798, and 2.524, isolating aggregate and tail control.

The independent trace-meter replay is a complementary deployment check: event-window MAE is 2.701/2.812/2.864 MW for feasible quantile/tail-risk counterfactual/risk-constrained verifieron the same 54 locked days (three-day block intervals in Experiment 20). It measures closed-DCGM alignment and remains separate from the mechanism-isolation response panel.

The closest domain baselines are equation-level structural controls. Spatial-DR, flexibility, receding-horizon, and all-site exact response give daily (nRMSE, false credit, F_1) of (3.063,114.572,0.137), (0.370,1.549,0.411), (0.345,1.849,0.442), and (0.329,0,0.279), respectively; DC–T/DC–ST translations give (0.370,1.549) with migration 0/4.328 MWh. Every control shares the ledger, windows, capacities, event slots, and locked days; the fairness audit records equation-level controls with no claim of software reimplementation [23], [33]. The receding-horizon translation follows [10]. Fig. 3 reports the distributions and ablations; its 810 schedules have binding shares 0.310/0.948 and nonzero slope 1.003 over 96–608 intervals.

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms. Table III isolates the mechanism with the trace-anchored reference and reports the locked workload rows. Exact signed nodal value has MAE \$151.719 versus \$360.580 for nodal gross response; the corresponding workload errors are \$455.865 and \$421.677. Signed netting reduces the trace error

TABLE III
BASE-CASE AND COMPLETE N–1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Uniform gross	360.528	421.230	–	–
Base case	Nodal gross	360.580	421.677	–	–
Base case	Uniform signed	150.928	454.651	–	–
Base case	Nodal signed linear	152.390	455.842	–	–
Base case	Nodal exact net value	**151.719**	455.865	–	–
N–1	Base exact	139.221	167.367	79.79/130.42	121.01/107.69
N–1	N–1 linear	2.208	**150.461**	0.60/1.91	113.43/20.38
N–1	**N–1 exact**	**0.990**	150.955	0.87/0.01	113.43/20.18

by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$), whereas uniform-to-nodal pricing changes it by $-\$1.463$ ($p = 0.0118$) and linear-to- exact value by \$0.671 ($p = 0.8026$). Workload effects are $-\$34.165$, $-\$1.191$, and $-\$0.023/day$, quantifying the mechanism-specific trade-offs on the same locked workload panel. All 5,616 interval/counterfactual instances satisfy (24); the largest violation is 2.03×10^{-11} dollars.

Experiment 4 applies the predeclared largest gross-minus-net offset (day 87), exposing the rebound component that positive-only local accounting omits.

C. Cross-network and N–1 security evidence

Experiment 5 evaluates 6,480 four-network outcomes; exact value improves the trace-anchored error in all 12 cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05), with a \$0.050/day workload change. Experiment 6 solves 810 capacity/deadline schedules (binding shares 0.031–0.911; false credit 2.374–2.740 MWh).

Experiment 7 enumerates 16 four-site and 256 eight-site coalitions (maximum budget residual 5.68×10^{-14} dollars) and 1,296 states in the exact 20-participant panel. Experiment 8 enforces (29) for all 37 RTS-24 non-islanding outages: trace/workload MAE is \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 for base exact/N–1 linear/N–1 exact, with 1.000 p.u. maximum loading.

Experiment 9 tests (31) with 54 global programs and a validation-frozen seven-profile hull. Held-out factors 0.734/0.996/1.278 have no cap violation above 10^{-8} USD; independent 40-segment MAE is \$181.266/\$169.652/\$177.013/\$169.652 per day for payment-certified verifier, single, feasible-quantile, and risk-constrained verifier. Experiment 15 gives interval widths \$11.229/\$22.054 at q_{01}/capacity-safe q_{99} (maxima \$50.090/\$97.951); $q_{99} = 2.6931$ is clipped to 1.150. The unseen-factor transfer panel (0.80/1.20) gives payment-certified MAE $40.914/61.567$ and feasible-quantile MAE $37.931/57.077$, with no factor used in certification.

Experiment 10 replaces the linear network with AC OPF [12]: 864 base and 5,400 corrective IEEE-30/39/118 outages satisfy the limits; 3,888 shared-active-plan cells have zero deviation (maximum loading 0.880/0.907/0.935 p.u. at 3%/6%/9%). Experiment 11 spans 24 regional permutations, three penetrations, four counterfactuals, and 54 days; RiskSafe/single MAE is \$105.250/\$211.555/\$315.207 and feasible-quantile is \$81.500/\$164.024/\$243.866 per day. Experiment 16 audits 66,769/28,413 training/test jobs (MAE 11.223 W, RMSE 15.818 W, $R^2 = 0.8994$), while Experiment 12 reports complete-cycle/event-window MAE \$0.641/\$0.658/\$0/\$0 and \$21.882/\$29.287/\$5.325/\$5.325; all 54 contracts activate.

Fig. 4 summarizes the cross-network preventive certificate: maximum apparent-power loading, voltage-limit deviation, and non-reference active-plan deviation for each penetration and public network.

D. Independent trace and deployment-boundary audits

Immutable MIT scheduler/DCGM replay gives slot MAE 3.740 versus 5.678 MW at 100% coverage; the 71,128-job

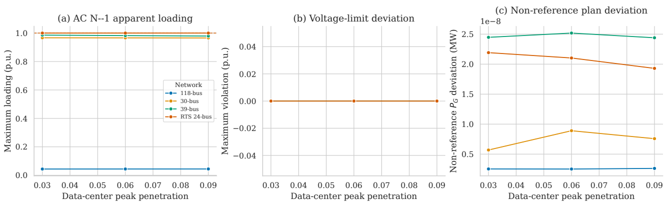

Fig. 4. Cross-network preventive AC N–1 certificate. Each point aggregates the predeclared snapshots and all native-case admissible connected outages; the dashed line in (a) is the unit apparent-power limit.

TABLE IV
EVENT-GATE INFORMATION BOUNDARY AND CONTRACT-CAPPED OPERATING REPLAY (54-DAY MEANS). THE SEPARATE TRACE-METER REPLAY IS OBSERVATIONAL AND HAS NO EVENT LABEL.

Method	nRMSE	Gross false MWh	Payable MWh	Meter underpay	Recall	F1
Gate-consistent submitted baseline	0.452	0	0	22.288	0	0
Committed rolling response	**0.358**	**3.442**	**7.015**	**17.919**	**0.292**	**0.283**
Committed-ledger reference	0.452	0	0	22.288	0	0
Complete-ledger RiskSafe	0.343	1.717	0	22.288	0.353	0.442

residual is below 10^{-17} MWh. The observational trace-meter panel gives event-window MAE 2.864 MW for risk-constrained verifier(2.701 MW for feasible quantile). With the slot-60 gate, the submitted baseline is frozen before scoring; the committed-ledger replay reaches nRMSE 0.358, gross false credit 3.442, meter-capped payable credit 7.015 MWh/day, recall 0.292, and $F_1 = 0.283$, while complete-ledger risk-constrained verifieris 0.343/1.717 with recall 0.353. The selected \$100/MWh, weight 15.0, and $\eta = 0.60$ reserve satisfy the 16-day, 7-MWh validation budget; Table IV keeps payable credit separate from offline diagnostics. Experiment 18 supplies 9,648 fixed-active-plan AC-OPF outcomes at the 0.90 native-load multiplier over three locked days and two slot endpoints: 32/35/24/177 admissible outages in RTS-24/IEEE-30/39/118, with maxima 1.000000 p.u. loading, zero voltage violation, 2.52×10^{-8} MW plan deviation, and minimum voltage 0.9503 p.u.

E. Indexed workload counterfactual

Experiment 19 assigns one service variable to each positive-energy job and admissible slot before its declared endpoint, retaining the 128-slot horizon, 0.001-MW/GPU bound, native region, exact energy, and 118-MW capacity. The sparse HiGHS program is preemptive; Experiment 14 is the contiguous witness. Table V reports native/counterfactual event energy 0.194/0.028 MWh, gross/net reduction 0.183/0.167 MWh, rebound 0.016 MWh, residual 2.17×10^{-19} MWh, and minimum slack 29.492 MWh. Exp. 21 certifies the $3475.108\times$ capacity-proportional homogeneous scale; with the 0.001-MW/GPU nameplate fixed, the deployable scale is $1.036\times$.

VI. CONCLUSION

This paper couples a provenance-bound workload verifier to signed nodal settlement. The safe profile reaches nRMSE/false credit 0.343/1.717 MWh/day; the one-hour gate uses distinct submitted-baseline and response LPs (Table IV). The indexed counterfactual preserves 71,128 job energies and yields 0.167 MWh net reduction (0.183 gross, 0.016 rebound). The scale certificate reports both $3475.108\times$ capacity-proportional and $1.036\times$ fixed-nameplate witnesses under the declared ledger mapping. Deployment requires signed co-located telemetry, topology access, and capacity-safe power conversion.

TABLE V
EXACT INDEXED JOB-LEVEL COUNTERFACTUAL CERTIFICATE.

Quantity	Value
Positive-energy jobs / job–slot variables	71,128 / 5,465,157
Native / counterfactual event energy (MWh)	0.194 / 0.028
Gross / net reduction (MWh)	**0.183 / 0.167**
Event rebound (MWh)	0.016
Maximum job-energy residual (MWh)	2.17×10^{-19}
Minimum regional-slot capacity slack (MWh)	29.492
Capacity-proportional homogeneous scale (×)	3475.108
Fixed 0.001-MW/GPU-nameplate scale (×)	1.036
Scale audit re-optimised	No

REFERENCES

[1] California Independent System Operator, "California iso baseline accuracy work group final proposal," tech. rep., California ISO, 2017.
[2] X. Wang and W. Tang, "Modeling and analysis of baseline manipulation in demand response programs," *IEEE Transactions on Smart Grid*, vol. 13, no. 2, pp. 1178–1186, 2022.
[3] Z. Liu, A. Wierman, Y. Chen, B. Razon, and N. Chen, "Data center demand response: Avoiding the coincident peak via workload shifting and local generation," *Performance Evaluation*, vol. 70, no. 10, pp. 770–791, 2013.
[4] M. A. Adnan, R. Sugihara, Y. Ma, and R. Gupta, "Energy efficient geographical load balancing via dynamic deferral of workload," in *2012 IEEE Fifth International Conference on Cloud Computing*, pp. 188–195, 2012.
[5] Y. Cao, F. Cao, Y. Wang, J. Wang, L. Wu, and Z. Ding, "Managing data center cluster as non-wire alternative: A case in balancing market," *Applied Energy*, vol. 360, p. 122769, 2024.
[6] I. Riepin, T. Brown, and V. Zavala, "Spatio-temporal load shifting for truly clean computing," *Advances in Applied Energy*, vol. 17, p. 100202, 2025.
[7] M. T. Takci, M. Qadrdan, J. Summers, and J. Gustafsson, "Data centres as a source of flexibility for power systems," *Energy Reports*, vol. 13, pp. 3661–3671, 2025.
[8] W. Zhang, L. A. Roald, A. A. Chien, J. R. Birge, and V. M. Zavala, "Flexibility from networks of data centers: A market clearing formulation with virtual links," *Electric Power Systems Research*, vol. 189, p. 106723, 2020.
[9] W. Zhang and V. M. Zavala, "Remunerating space–time, load-shifting flexibility from data centers in electricity markets," *Applied Energy*, vol. 326, p. 119930, 2022.
[10] W. Zhang, L. A. Roald, and V. M. Zavala, "Exploring the impacts of power grid signals on data center operations using a receding-horizon scheduling model," *IEEE Transactions on Sustainable Computing*, vol. 8, no. 2, pp. 245–256, 2023.
[11] B. Stott, J. Jardim, and O. Alsac, "DC power flow revisited," *IEEE Transactions on Power Systems*, vol. 24, no. 3, pp. 1290–1300, 2009.
[12] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
[13] D. A. Tejada-Arango, P. Sánchez-Martín, and A. Ramos, "Security constrained unit commitment using line outage distribution factors," *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 329–337, 2018.
[14] National Institute of Standards and Technology and Q. Dang, "Secure hash standard," Tech. Rep. FIPS PUB 180-4, National Institute of Standards and Technology, 2015.
[15] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.
[16] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
[17] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
[18] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Machine Learning*, vol. 63, pp. 3–42, 2006.
[19] A. Abadie, A. Diamond, and J. Hainmueller, "Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program," *Journal of the American Statistical Association*, vol. 105, no. 490, pp. 493–505, 2010.
[20] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.
[21] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018.
[22] B. Satchidanandan, M. Roozbehani, and M. A. Dahleh, "A two-stage mechanism for demand response markets," *IEEE Control Systems Letters*, vol. 7, pp. 49–54, 2023.
[23] M. Chen, C. Gao, M. Shahidehpour, and Z. Li, "Incentive-compatible demand response for spatially coupled internet data centers in electricity markets," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3056–3069, 2021.
[24] J. John F. Nash, "The bargaining problem," *Econometrica*, vol. 18, no. 2, pp. 155–162, 1950.
[25] North American Electric Reliability Corporation, "TPL-001-5.1: Transmission system planning performance requirements," tech. rep., North American Electric Reliability Corporation, 2020.
[26] S. Samsi, M. L. Weiss, D. Bestor, B. Li, M. Jones, A. Reuther, *et al.*, "The mit supercloud dataset," in *2021 IEEE High Performance Extreme Computing Conference*, pp. 1–8, 2021.
[27] L. S. Shapley, "A value for n-person games," in *Contributions to the Theory of Games, Volume II* (H. W. Kuhn and A. W. Tucker, eds.), pp. 307–317, Princeton University Press, 1953.
[28] Y. Wang, Y. Chen, Z. Li, X. Kang, Y. Fang, Y. Zhou, Y. Zheng, Z. Tang, X. He, R. Guo, X. Wang, Q. Wang, A. C. Zhou, and X. Chu, "BurstGPT: A real-world workload dataset to optimize large language model serving systems," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 5831–5841, 2025.
[29] S. Babaeinejadsarookolaee, A. Birchfield, R. D. Christie, C. Coffrin, C. DeMarco, R. Diao, *et al.*, "The power grid library for benchmarking ac optimal power flow algorithms," *arXiv preprint arXiv:1908.02788*, 2019.
[30] C. Grigg, P. Wong, P. Albrecht, R. Allan, M. Bhavaraju, R. Billinton, Q. Chen, C. Fong, S. Haddad, S. Kuruganty, *et al.*, "The IEEE reliability test system–1996," *IEEE Transactions on Power Systems*, vol. 14, no. 3, pp. 1010–1020, 1999.
[31] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.
[32] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
[33] Y. Cao, M. Cheng, S. Zhang, H. Mao, P. Wang, C. Li, Y. Feng, and Z. Ding, "Data-driven flexibility assessment for internet data center towards periodic batch workloads," *Applied Energy*, vol. 324, p. 119665, 2022.